

Test Protocol — Killer Sudoku Web Application

Project: Killer Sudoku (Skills Battle 2026 — Application Development) **Author:** Korel Uyar **Date:** 2026-05-27 **Document version:** INITIAL (test plan only; "Actual Result" and "Status" columns are filled in the FINAL revision after implementation)

1. Scope

This document lists every test case that will be executed to verify the Killer Sudoku web application against the 14 use cases (11 mandatory + 3 additional).

For every use case, the protocol covers at least one **positive**, one **negative**, and where meaningful one **boundary** test. Pure-logic cases (solver, validator, hint, scoring) are additionally implemented as **automated unit tests** with Vitest; the column "Unit Test?" marks them with Y. UI-driven cases are executed manually.

Legend for the **Status** column (filled in FINAL revision only):

Symbol	Meaning
✔	Pass
✖	Fail
⚠	Pass with caveat (see notes)
🕒	Not yet executed

2. Test Environment

- Node.js 20+, MySQL 8+
- Browser: Chromium 120+ (manual cases)
- Tests run with `npm test` (Vitest)
- Database seeded with admin user `admin / Admin1234` and 9 puzzles (3 per difficulty)

3. Test Cases

#	Use Case	Test Case Name	Type	Unit Test?	Preconditions	Steps	Expected Result	Actual Result	Status
1	UC1: read rules	Guest opens landing page	Positive	N	App running, not logged in	Visit <code>/</code>	Hero, rules teaser and "Read full rules" link visible	TBD	🕒
2	UC1: read rules	Full rules page renders example	Positive	N	App running	Visit <code>/rules</code>	Full rules + animated example Killer Sudoku puzzle visible	TBD	🕒
3	UC1: read rules	Rules accessible without auth	Negative	N	Logged out	Open <code>/rules</code> directly via URL	200 OK, no redirect to login	TBD	🕒
4	UC1: read rules	Rules link from navbar always visible	Positive	N	Any auth state	Inspect navbar on every page	"Rules" link present	TBD	🕒
5	UC2: create user	Register with valid credentials	Positive	Y	DB clean	POST <code>/api/auth/register</code> with <code>{username:"alice", email:"a@b.de", password:"Pass1234"}</code>	201 Created, user row in DB with bcrypt hash	TBD	🕒
6	UC2: create user	Duplicate username rejected	Negative	Y	User "alice" exists	POST register with username "alice" again	409 Conflict, no row inserted	TBD	🕒

#	Use Case	Test Case Name	Type	Unit Test?	Preconditions	Steps	Expected Result	Actual Result	Status
7	UC2: create user	Invalid email format rejected	Negative	Y	DB clean	POST register with email "not-an-email"	400 Bad Request, Zod error	TBD	🏆
8	UC2: create user	Username with 3 chars accepted (min)	Boundary	Y	DB clean	POST register username="abc"	201 Created	TBD	🏆
9	UC2: create user	Username with 20 chars accepted (max)	Boundary	Y	DB clean	POST register with 20-char username	201 Created	TBD	🏆
10	UC2: create user	Username 21 chars rejected	Boundary	Y	DB clean	POST register with 21-char username	400 Bad Request	TBD	🏆
11	UC2: create user	Password without digit rejected	Negative	Y	DB clean	POST register password="onlyletters"	400 Bad Request	TBD	🏆
12	UC2: create user	Password under 8 chars rejected	Boundary	Y	DB clean	POST register password="A1b2c3"	400 Bad Request	TBD	🏆
13	UC3: login	Login with valid creds returns JWT cookie	Positive	Y	User exists	POST <code>/api/auth/login</code> valid creds	200 OK, HttpOnly cookie <code>auth_token</code> set	TBD	🏆
14	UC3: login	Wrong password rejected	Negative	Y	User exists	POST login wrong password	401 Unauthorized, no cookie	TBD	🏆
15	UC3: login	Non-existent user rejected	Negative	Y	DB clean	POST login username "ghost"	401 Unauthorized	TBD	🏆
16	UC3: login	Authenticated <code>/api/auth/me</code> returns user	Positive	N	Logged in	GET <code>/api/auth/me</code>	200 OK with <code>{id, username, email}</code>	TBD	🏆
17	UC3: login	Logout clears cookie	Positive	N	Logged in	POST <code>/api/auth/logout</code>	200 OK, cookie cleared, subsequent <code>/me</code> → 401	TBD	🏆
18	UC4: enter puzzle	Logged-in user opens <code>/create</code>	Positive	N	Logged in	Visit <code>/create</code>	Empty 9x9 grid + cage drawing tools shown	TBD	🏆
19	UC4: enter puzzle	Guest redirected from <code>/create</code>	Negative	N	Not logged in	Visit <code>/create</code>	Redirect to <code>/auth/login</code>	TBD	🏆
20	UC4: enter puzzle	Cage of 1 cell allowed	Boundary	Y	Builder open	Draw cage covering single cell, set sum=5	Cage saved client-side	TBD	🏆
21	UC4: enter puzzle	Cage of 9 cells allowed	Boundary	Y	Builder open	Draw cage covering an entire row	Cage saved	TBD	🏆
22	UC4: enter puzzle	Overlapping cages rejected	Negative	Y	Builder open	Attempt to draw a cage over already-caged cells	Error toast, cage not saved	TBD	🏆
23	UC4: enter puzzle	Cells not in any cage prevents save	Negative	Y	Builder open	Click Save with 80 cells caged	Error: "All 81 cells must be inside a cage"	TBD	🏆
24	UC4: enter puzzle	Difficulty selector enforces 1–3	Boundary	Y	Builder open	Try to set difficulty=4	UI prevents value, only 1/2/3 selectable	TBD	🏆

#	Use Case	Test Case Name	Type	Unit Test?	Preconditions	Steps	Expected Result	Actual Result	Status
25	UC5: save new puzzle	Save valid puzzle persists in DB	Positive	Y	Logged in, valid puzzle in builder	POST <code>/api/puzzles</code>	201 Created, row in <code>puzzles</code> table	TBD	🏆
26	UC5: save new puzzle	Reject puzzle where Σ cages $\neq$ 405	Negative	Y	Puzzle with cage sums totalling 400	POST <code>/api/puzzles</code>	400 Bad Request, error "Cage sums must total 405"	TBD	🏆
27	UC5: save new puzzle	Reject unsolvable puzzle	Negative	Y	Puzzle constructed so no assignment works	POST <code>/api/puzzles</code>	400 Bad Request, error "Puzzle has no solution"	TBD	🏆
28	UC5: save new puzzle	Reject puzzle with multiple solutions	Negative	Y	Puzzle with ambiguous cages	POST <code>/api/puzzles</code>	400 Bad Request, error "Solution is not unique"	TBD	🏆
29	UC5: save new puzzle	Cage sum 45 for full-row cage accepted	Boundary	Y	Puzzle with one 9-cell cage summing 45 + rest of cages valid	POST <code>/api/puzzles</code>	201 Created	TBD	🏆
30	UC6: solve puzzle	Open existing puzzle renders grid	Positive	N	Puzzle id 1 exists	Visit <code>/play/1</code>	Grid + cages render correctly	TBD	🏆
31	UC6: solve puzzle	Non-existent puzzle returns 404	Negative	N	id 99999 not in DB	Visit <code>/play/99999</code>	404 page	TBD	🏆
32	UC6: solve puzzle	Number entered persists in client state	Positive	N	Puzzle open	Click cell, press 5	Cell shows 5, value retained in Zustand store	TBD	🏆
33	UC6: solve puzzle	Notes mode places candidates	Positive	N	Puzzle open	Toggle Notes, click cell, press 3 then 7	3 and 7 shown as small candidates in cell	TBD	🏆
34	UC6: solve puzzle	Arrow keys navigate grid	Positive	N	Cell selected	Press ArrowRight, ArrowDown	Selection moves accordingly	TBD	🏆
35	UC7: ask for hint	Hint reveals correct value	Positive	Y	Puzzle in progress	POST <code>/api/puzzles/:id/hint</code> with current state	200 OK, returns <code>{row, col, value}</code> matching unique solution	TBD	🏆
36	UC7: ask for hint	Hint counter increments	Positive	Y	hintsUsed=0 in state	Request hint	Store hintsUsed=1, UI badge updates	TBD	🏆
37	UC7: ask for hint	Hint on already-solved grid is no-op	Negative	Y	All cells filled correctly	Request hint	200 OK with <code>null</code> , counter NOT incremented	TBD	🏆
38	UC7: ask for hint	Hint picks MRV cell	Positive	Y	State where one empty cell has 1 candidate	Request hint	Returned cell is the MRV cell	TBD	🏆
39	UC8: show high score	Leaderboard sorted by score ascending	Positive	N	3 results in DB	GET <code>/api/leaderboard</code>	Results ordered: best (lowest score) first	TBD	🏆
40	UC8: show high score	Filter by difficulty=2	Positive	N	Results across difficulties	GET <code>/api/leaderboard?difficulty=2</code>	Only difficulty=2 results	TBD	🏆

#	Use Case	Test Case Name	Type	Unit Test?	Preconditions	Steps	Expected Result	Actual Result	Status
41	UC8: show high score	Empty leaderboard renders empty state	Boundary	N	No results	Visit <code>/leaderboard</code>	"No results yet" message	TBD	🏆
42	UC8: show high score	Score = timeSeconds + hintsUsed × 60	Positive	Y	Result(time=300, hints=2)	Compute score	Score = 420	TBD	🏆
43	UC9: check solution	Valid full grid passes	Positive	Y	All cells correct	POST <code>/api/puzzles/:id/check</code>	200 OK with <code>{ok: true}</code>	TBD	🏆
44	UC9: check solution	Invalid solution rejected	Negative	Y	One cell wrong	POST check	200 OK with <code>{ok: false, conflicts: [...]}</code>	TBD	🏆
45	UC9: check solution	Incomplete grid rejected	Negative	Y	Some cells empty	POST check	200 OK with <code>{ok: false, reason: "incomplete"}</code>	TBD	🏆
46	UC9: check solution	Cage sum mismatch reported	Negative	Y	Full grid but cage sum wrong	POST check	<code>{ok: false, reason: "cage_sum"}</code> with cage id	TBD	🏆
47	UC10: save result	Authenticated save persists	Positive	N	Logged in, puzzle solved	POST <code>/api/results {puzzleId, timeSeconds:300, hintsUsed:1}</code>	201 Created, row in <code>results</code>	TBD	🏆
48	UC10: save result	Guest save rejected	Negative	N	Not logged in	POST <code>/api/results</code>	401 Unauthorized	TBD	🏆
49	UC10: save result	Result with 0 hints accepted	Boundary	N	Logged in	POST result with hintsUsed=0	201 Created	TBD	🏆
50	UC10: save result	Negative time rejected	Negative	Y	Logged in	POST result with timeSeconds=-1	400 Bad Request	TBD	🏆
51	UC11: auto solve	Solver solves a valid puzzle within 2s	Positive	Y	Sample medium puzzle	Call <code>solve(grid, cages)</code>	Returns full solution; duration < 2000ms	TBD	🏆
52	UC11: auto solve	Solver returns null on unsolvable	Negative	Y	Constructed unsolvable puzzle	Call <code>solve()</code>	Returns <code>null</code>	TBD	🏆
53	UC11: auto solve	Solver handles single empty cell	Boundary	Y	Grid with 80/81 filled correctly	Call <code>solve()</code>	Returns grid with last cell filled	TBD	🏆
54	UC11: auto solve	Uniqueness check detects single solution	Positive	Y	Standard puzzle	Call <code>countSolutions(grid, cages, max=2)</code>	Returns 1	TBD	🏆
55	UC11: auto solve	Uniqueness check detects multiple	Negative	Y	Ambiguous puzzle	Call <code>countSolutions(grid, cages, max=2)</code>	Returns 2	TBD	🏆
56	UC11: auto solve	Σ cages ≠ 405 short-circuits before backtracking	Positive	Y	Puzzle with cage sums = 400	Call <code>validate(cages)</code>	Returns false immediately, no solver invoked	TBD	🏆
57	UC12: daily	<code>/daily</code> shows today's puzzle	Positive	N	Daily puzzle for today exists	Visit <code>/daily</code>	Grid for today's daily renders	TBD	🏆

#	Use Case	Test Case Name	Type	Unit Test?	Preconditions	Steps	Expected Result	Actual Result	Status
58	UC12: daily	Same daily for all users on same date	Positive	Y	Two users, same date	Both call GET /api/daily	Both receive same puzzleId	TBD	🏆
59	UC12: daily	Daily leaderboard scoped to today	Positive	N	Results on previous days exist	GET /api/daily/leaderboard	Only today's results returned	TBD	🏆
60	UC12: daily	First request of day creates daily row	Boundary	Y	No daily for today yet	GET /api/daily	Row inserted, daily puzzle id returned	TBD	🏆
61	UC13: rate puzzle	Rate puzzle after solve	Positive	N	Result saved for puzzle 1	POST /api/ratings {puzzleId:1, stars:5, difficultyFeedback:"fits"}	201 Created	TBD	🏆
62	UC13: rate puzzle	Cannot rate without solving	Negative	N	No result for user/puzzle	POST rating	403 Forbidden	TBD	🏆
63	UC13: rate puzzle	Stars outside 1–5 rejected	Negative	Y	Logged in	POST rating stars=6	400 Bad Request	TBD	🏆
64	UC13: rate puzzle	Second rating updates first (upsert)	Positive	Y	Existing rating for user/puzzle	POST rating again with different stars	200 OK, row updated not duplicated	TBD	🏆
65	UC13: rate puzzle	Average rating shown on puzzle card	Positive	N	3 ratings: 4, 5, 3	GET /api/puzzles	Average 4.0 displayed	TBD	🏆
66	UC14: stats	Stats page shows solved count	Positive	N	User solved 5 puzzles	Visit /stats	"Solved: 5" displayed	TBD	🏆
67	UC14: stats	Best time per difficulty	Positive	Y	Multiple results	GET /api/stats	Per-difficulty best time correct	TBD	🏆
68	UC14: stats	Empty stats for new user	Boundary	N	New user, no results	Visit /stats	Empty-state with CTA "Solve your first puzzle"	TBD	🏆
69	UC14: stats	Win streak counts consecutive days	Positive	Y	Results on 3 consecutive days	Compute streak	Streak = 3	TBD	🏆
70	UC14: stats	Streak resets on missed day	Boundary	Y	Results day 1, 2, skip 3, 4	Compute streak	Current streak = 1	TBD	🏆
71	UC3: login	Navbar updates immediately after sign-in (no manual refresh)	Positive	N	Logged out, on /auth/login	Submit valid creds	Top-right navbar shows the avatar + username without the user having to refresh the browser	TBD	🏆
72	UC7: ask for hint	Hint adds a +60s penalty to the displayed timer	Positive	N	Puzzle in progress, timer running	Click the Hint button	Displayed timer immediately jumps forward by 60 seconds; a small "+60s" indicator floats above the timer	TBD	🏆
73	UC7: ask for hint	Hint corrects a wrong user value, not only empty cells	Positive	Y	Player has typed a wrong digit into at least one cell	Click Hint	Hint API returns a cell whose user-value differs from the canonical solution, and provides the correct value; the system does not wrongly say "grid is complete"	TBD	🏆

4. Coverage Summary

Use Case	Total	Positive	Negative	Boundary	Unit Tests
UC1 read rules	4	3	1	0	0
UC2 create user	8	1	4	3	8
UC3 login	6	4	2	0	3
UC4 enter puzzle	7	1	3	3	5
UC5 save new puzzle	5	1	3	1	5
UC6 solve puzzle	5	4	1	0	0
UC7 ask for hint	6	5	1	0	5
UC8 high score	4	2	0	1	1
UC9 check solution	4	1	3	0	4
UC10 save result	4	1	2	1	1
UC11 auto solve	6	4	2	1	6
UC12 daily	4	3	0	1	2
UC13 rate puzzle	5	3	2	0	2
UC14 stats	5	3	0	2	3
Total	73	36	24	13	45

73 total test cases (well above the 50 minimum), 45 of which are automated via Vitest.

5. Execution Procedure

1. Run `npm install` to install dependencies.
2. Provision MySQL and apply `sudoku.sql`, then run `npx prisma db push`.
3. Seed with `npx tsx seed.ts` — this creates the `admin` user, 9 puzzles, today's daily entry.
4. Run `npm test -- --run` for the 45 unit tests.
5. Run `npm run dev` and execute the manual cases against `http://localhost:3000`.
6. Record outcomes in this document's *Actual Result* / *Status* columns and re-export the FINAL PDF.